INCH-POUND
MS19068C
w/AMENDMENT 1
07 August 2026
SUPERSEDING
MS19068C
14 March 2023

DETAIL SPECIFICATION SHEET

NUTS, PLAIN, ROUND, RETAINING, BALL AND ROLLER BEARING, REGULAR SERIES

This specification is approved for all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and procurement specification MIL-DTL-21337.

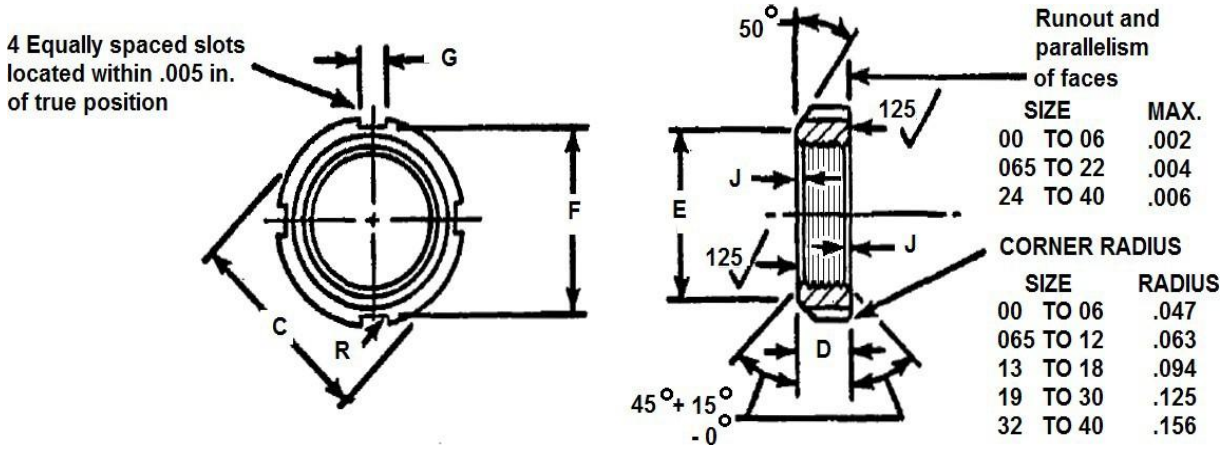


FIGURE 1. Nut.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil

AMSC N/A FSC 5310

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:20Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and dimensions

Dash No.	D Thickness		C 0D +0.005 -0.015	E Face Dim		J Cham	Slot dimensions F +0.010 -0.020	G		R	Thread data No. per inch	Minor dia		Pitch dia		Major dia
	Min	Max		Min	Max			Min	Max	Max		Min	Max	Min	Max	Min
00	0.209	0.229	0.750	0.605	0.625	0.031	0.625	0.120	0.130	0.010	32	0.3572	0.3606	0.3707	0.3733	0.391
01	0.303	0.323	0.875	0.699	0.719	0.031	0.750	0.120	0.130	0.010	32	0.4352	0.4386	0.4487	0.4513	0.469
02	0.303	0.323	1.000	0.793	0.813	0.031	0.812	0.120	0.130	0.010	32	0.5522	0.5556	0.5657	0.5687	0.586
03	0.334	0.354	1.125	0.918	0.938	0.031	0.937	0.120	0.130	0.010	32	0.6302	0.6336	0.6437	0.6467	0.664
04	0.365	0.385	1.375	1.105	1.125	0.031	1.187	0.178	0.198	0.015	32	0.7472	0.7506	0.7607	0.7641	0.781
05	0.396	0.416	1.562	1.261	1.281	0.031	1.375	0.178	0.198	0.015	32	0.9352	0.9386	0.9487	0.9521	0.969
06	0.396	0.416	1.750	1.480	1.500	0.047	1.562	0.178	0.198	0.015	18	1.1129	1.1189	1.1369	1.1409	1.173
065	0.428	0.448	2.062	1.793	1.813	0.047	1.875	0.178	0.198	0.015	18	1.2524	1.2584	1.2764	1.2804	1.313
07	0.428	0.448	2.062	1.793	1.813	0.047	1.875	0.178	0.198	0.015	18	1.3159	1.3219	1.3399	1.3439	1.376
08	0.428	0.448	2.250	1.980	2.000	0.047	2.062	0.240	0.260	0.020	18	1.5029	1.5089	1.5269	1.5314	1.563
09	0.428	0.448	2.531	2.261	2.281	0.047	2.343	0.240	0.260	0.020	18	1.7069	1.7129	1.7309	1.7354	1.767
10	0.49	0.51	2.687	2.418	2.438	0.047	2.500	0.240	0.260	0.020	18	1.9069	1.9129	1.9309	1.9354	1.967
11	0.49	0.51	2.968	2.636	2.656	0.047	2.718	0.240	0.260	0.020	18	2.0969	2.1029	2.1209	2.1260	2.157
12	0.521	0.541	3.156	2.824	2.844	0.047	2.906	0.240	0.260	0.020	18	2.2999	2.3059	2.3239	2.3290	2.360
13	0.553	0.573	3.375	3.043	3.063	0.047	3.125	0.240	0.260	0.020	18	2.4879	2.4939	2.5119	2.5170	2.548
14	0.553	0.573	3.625	3.283	3.313	0.047	3.375	0.240	0.260	0.020	18	2.6909	2.6969	2.7149	2.7200	2.751
15	0.548	0.0604	3.875	3.533	3.563	0.078	3.625	0.360	0.385	0.025	12	2.8428	2.8518	2.8789	2.8843	2.933
16	0.584	0.604	4.156	3.814	3.844	0.078	3.906	0.360	0.385	0.025	12	3.0468	3.0558	3.0829	3.0888	3.137
17	0.615	0.635	4.406	4.001	4.031	0.078	4.093	0.360	0.385	0.025	12	3.2498	3.2588	3.2859	3.2918	3.340
18	0.678	0.698	4.656	4.251	4.281	0.078	4.343	0.360	0.385	0.025	12	3.4368	3.4458	3.4729	3.4803	3.527
19	0.709	0.729	4.937	4.533	4.563	0.078	4.625	0.360	0.385	0.025	12	3.6398	3.6488	3.6759	3.6833	3.730
20	0.735	0.76	5.187	4.783	4.813	0.078	4.875	0.360	0.385	0.025	12	3.8278	3.8368	3.8639	3.8713	3.918
21	0.735	0.76	5.437	4.970	5.000	0.078	5.062	0.485	0.510	0.030	12	4.0318	4.0408	4.0679	4.0762	4.122
22	0.766	0.791	5.718	5.251	5.281	0.078	5.343	0.485	0.510	0.030	12	4.2348	4.2438	4.2709	4.2792	4.325
24	0.798	0.823	6.125	5.658	5.688	0.078	5.750	0.485	0.510	0.030	12	4.6258	4.6348	4.6619	4.6702	4.716
26	0.860	0.885	6.750	6.158	6.188	0.078	6.250	0.610	0.635	0.030	12	5.0158	5.0248	5.0519	5.0602	5.106
28	0.923	0.948	7.093	6.501	6.531	0.078	6.593	0.610	0.635	0.030	12	5.4068	5.4158	5.4429	5.4512	5.497
30	0.954	0.979	7.687	7.033	7.063	0.078	7.125	0.610	0.635	0.030	12	5.7978	5.8068	5.8339	5.8422	5.888
32	1.016	1.041	8.062	7.398	7.438	0.094	7.500	0.610	0.635	0.030	8	6.1487	6.1622	6.2028	6.2119	6.284
34	1.048	1.073	8.656	7.991	8.031	0.094	8.093	0.610	0.635	0.030	8	6.5237	6.5372	6.5778	6.5869	6.659
36	1.079	1.104	9.062	8.335	8.375	0.094	8.437	0.735	0.760	0.030	8	6.9307	6.9442	6.9848	6.9939	7.066
38	1.110	1.135	9.468	8.741	8.781	0.094	8.843	0.735	0.760	0.030	8	7.3367	7.3502	7.3908	7.3999	7.472
40	1.173	1.198	9.843	9.116	9.156	0.094	9.218	0.735	0.760	0.030	8	7.7117	7.7252	7.7658	7.7772	7.847

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:20Z
Check the source to verify that this is the current version before use.

NOTES:

1. MATERIALS:

   (a) Carbon steel, SAE 1015 to 1035 (UNS G10150 to UNS G10350), 1112 to 1118 (UNS G11120 to UNS G11180) or 1212 to 1215 (UNS G12120 to UNS G12150) in accordance with SAE J403.

   (b) Corrosion resisting steel: 302 (UNS S30200), 303 (UNS S30300) or 304 (UNS S30400) in accordance with ASTM A959.

2. HARDNESS:

   (a) Carbon steel nuts shall have a hardness of Rockwell HRB 66 to 92.

   (b) Corrosion resisting steel shall have a maximum hardness of Rockwell HRB 90.

3. PROTECTIVE COATING:

   (a) Carbon steel nuts are available either zinc plated or unplated. Zinc plated nuts shall be plated in accordance with ASTM B633, Class Fe/Zn 5, Type II. Cadmium plating in accordance with SAE AMS-QQ-P-416, Type II, Class 3 shall no longer be furnished.

   As a hexavalent chrome-free option, carbon steel nuts shall zinc plated in accordance with ASTM B633, Class Fe/Zn 5, Type V or VI, or zinc-nickel plated in accordance with MIL-PRF-32647, Type K20.

   (b) Corrosion resisting steel nuts shall be passivated in accordance with SAE AMS2700.

   As a hexavalent chrome-free option, corrosion resisting steel nuts shall be passivated in accordance with SAE AMS2700, Method 1, Type 6, Type 7 or Type 8, or Method 2.

4. TESTING: MIL-STD-1916, Test Method Standard, shall be utilized to test the fasteners for quality and inspection frequency.

5. THREADS: Threads shall be American National Form, Class UNC-3B, in accordance with FED-STD-H28.

6. SURFACE ROUGHNESS: Surface roughness shall be in accordance with ASME B46.1.

7. MAGNETIC PARTICLE INSPECTION: Inspection shall be in accordance with MIL-DTL-21337.

8. LIQUID PENETRANT INSPECTION: Inspection for corrosion resisting steel parts shall be in accordance with ASTM E1417/E1417M.

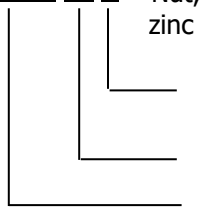
9. DIMENSIONS: All dimensions are in inches. Tolerances; decimals ± 0.010, angles ± 2º unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:20Z
Check the source to verify that this is the current version before use.

10. PART NUMBER: The MS19068 part number consists of the MS number, plus the size and the material number. The material numbers are:

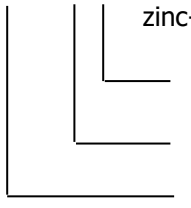
    1 – unplated carbon steel
    2 – zinc plated carbon steel
    3 – passivated corrosion resisting steel
    4 – zinc-nickel plated carbon steel
    5 – zinc plated carbon steel (hex chrome-free)
    6 – passivated corrosion resisting steel (hex chrome-free)

    Examples: MS19068-00 2 = Nut, Plain, Round, Retaining, Slot Dimension F = .625, zinc plated carbon steel

    Material number (1, 2, 3, 4, 5 or 6)
    Dash number per Table I
    Basic MS number

    MS19068-05 4 = Nut Plain, Round, Retaining, Slot Dimension F = 1.375, zinc-nickel plated carbon steel

    Material number (1, 2, 3, 4, 5 or 6)
    Dash number per Table I
    Basic MS number

11. WASHER: Use MS19070 for applicable key retaining washers. Applicable washers shall have size numbers corresponding to nut size numbers: Example: nut size 00 should be used with washer size number 00.

12. For design feature purpose, this specification takes precedence over procurement documents referenced herein.

13. Referenced documents shall be of the issue in effect on the date of invitations for bids, or request for proposal, except that referenced adopted industry documents shall give the date of the issue adopted.

14. Amendment notations. The margins of this document are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - AT
Navy – OS
DLA – GS

Preparing activity:
Army – AT

(Project 5310-2026-011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:20Z
Check the source to verify that this is the current version before use.